Project Initialization and Planning Phase

Date	26 June 2026
Team ID	N/A
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution is an AI-powered Voice Based Concept Understanding Analyser that evaluates a user's conceptual understanding from spoken responses. It converts speech into text, analyzes semantic similarity with reference concepts, measures speech quality, and generates a detailed performance report with personalized feedback.

Project Overview	
Objective	To develop an AI-based system that evaluates users' conceptual understanding through voice responses and provides accurate scores and feedback.
Scope	The system accepts voice input, converts speech to text, performs NLP and speech analysis, compares responses with reference concepts, and generates an evaluation report.
Problem Statement	
Description	Traditional oral assessments are subjective, time-consuming, and inconsistent. Manual evaluation makes it difficult to provide instant and unbiased feedback.
Impact	Automated evaluation improves assessment accuracy, saves evaluator time, provides instant feedback, and enhances learning outcomes.

Proposed Solution	
Approach	Develop a web-based application using Speech-to-Text, Natural Language Processing (NLP), semantic similarity analysis, and audio feature extraction to evaluate spoken responses.
Key Features	• Voice recording and upload. • Automatic speech-to-text conversion. • Concept similarity analysis using NLP. • Speech quality analysis (pause, filler words, fluency). • Understanding level prediction (Strong/Moderate/Poor). • Detailed report generation with feedback.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	4 Core CPU / Optional GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	2 GB SSD
Software		
Frameworks	Backend Framework	Flask
Libraries	Python Libraries	SpeechRecognition, Whisper, spaCy, Sentence Transformers, Librosa, NumPy, Pandas
Development Environment	IDE	VS Code, Jupyter Notebook
Data		

Data	Source, size, format (Voice recordings and reference concepts Custom)	Custom voice dataset (WAV/MP3) with concept reference text